Practice Quiz: Systems (Fiber & Flow)

Part A: Multiple Choice

1. In Active Inference, a Markov blanket is best described as: A) The plastic wrapper around a skein of yarn B) A statistical boundary separating internal states from external states C) The label on a yarn skein listing fiber content D) The pattern instructions for a crochet project

2. Which of the following is an **internal state** of a yarn system? A) The humidity in the room where the yarn is stored B) The fiber composition (wool, cotton, silk) C) The temperature of the crocheter's hands D) The lighting in the craft store

3. Two skeins from different dye lots differ because: A) They are made from completely different fibers B) The manufacturing environment varied slightly between batches C) One is always lighter and one is always darker D) Dye lots are arbitrary marketing labels with no real meaning

4. How does a tightly twisted yarn differ from a loosely twisted yarn as a system? A) Tight twist has fewer internal states B) Tight twist resists splitting and produces crisper stitches; loose twist is softer but splits more easily C) Tight twist is always better quality D) Twist direction has no effect on yarn behavior

5. When a crocheter squeezes a skein to evaluate it, they are: A) Damaging the yarn's internal structure B) Reading sensory information from the system boundary to infer internal states C) Performing a purely aesthetic judgment D) Testing whether the yarn will fit in their project bag

6. Blocking a finished piece works by: A) Using environmental factors (water, heat) to change the yarn system's internal states B) Adding new fibers to the yarn C) Removing the Markov blanket from the system D) Replacing the yarn's dye with a new color

7. Wool is hygroscopic, meaning it: A) Repels all moisture from its boundary B) Absorbs moisture from the environment, changing its weight and feel C) Changes color when exposed to water D) Cannot be used in humid climates

Part B: Short Answer

1. Explain in 2-3 sentences why gauge can change between a dry winter day and a humid summer day, even when the crocheter uses the same yarn, hook, and tension. Use the concept of environmental states affecting the yarn system.

2. Give a concrete example of how the yarn's active states (its outward influence on the environment) are visible when you work with mohair versus a smooth cotton.

3. A friend says "yarn is just string — it doesn't have states or boundaries." Using what you learned in this module, write a brief response that explains why yarn genuinely functions as a system with a Markov blanket.